



LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING, (AUTONOMOUS)
Approved by AICTE, Affiliated to JNTUK, Accredited by NBA,
ISO 9001:2008 Certified & Accredited by NAAC with "A" Grade
EXAMINATION SECTION

M.TECH. I SEMESTER REGULAR AND SUPPLEMENTARY END EXAMINATIONS, MARCH/APRIL 2015

TIME : 10.00 AM To 01.00 PM						
Date	Regular/ Supplementary	(CSE) Computer Science and Engineering	(IT) Software Engineering	(ECE) Systems and Signal Processing	(ME) Thermal Engineering	(PED) Power Electronics and Drives
30-03-2015 Monday	Regular	MTCS101 - Advanced Data Structures ✓	MTIT101- Advanced Problem Solving ✓	MTEC101 - Advanced Digital Signal Processing ✓	MTNE101 - Mathematical Methods in Engineering	MTEE101- Machine Modeling and Analysis ✓
	Supplementary	MCS101-Data Structures & Algorithm Analysis 2	MIT101-Requirements Engg. & Estimation ✓	MEC101-Advanced Digital Signal Processing ✓	MME101-Advanced Mathematics 2	MEE101-Machine Modeling and Analysis ✓
04-04-2015 Wednesday	Regular	MTCS102 - Advanced Database Management Systems ✓	MTIT102 - Software Architecture ✓	MTEC102- Digital Signal Processors ✓	MTNE102- Advanced Thermodynamics	MTEE102- Power Converters ✓
	Supplementary	MCS102-Database Management Systems 2	MIT102-Software Architectures ✓	MEC102-Transform Techniques ① 2	MME102-Advanced Thermodynamics 2	MEE102-Power Converters ① 2
04-04-2015 Saturday	Regular	MTCS103 - Data Mining ✓	MTIT103- Software Process Management ✓	MTEC103- Linear Algebra for Signal Processing ✓	MTNE103 - Advanced Heat & Mass Transfer	MTEE103 - Control of Motor Drives-I ✓
	Supplementary	MCS103-Computer Organization 2	MIT103-Software Process Management ✓	MEC103-VLSI Technology and Design ①	MME103-Advanced Heat & Mass Transfer 2	MEE103-Control of Motor Drives-I ✓
06-04-2015 Monday	Regular	MTCS104 - Advanced Computer Networks ✓	MTIT104- Object Oriented Software Engineering ✓	MTEC104- Soft Computing Techniques ✓	MTNE104- Advanced IC Engines ✓	MTEE104 - Computational Mathematics ✓
	Supplementary	MCS104-Computer Networks ✓	MIT104-Web Searching and Mining ✓	MEC104-Microcontrollers For Embedded System Design ①	MME104-IC Engine Combustion & Pollution ✓	MEE104-Special machines ✓
08-04-2015 Tuesday	Regular	Elective-I MTCS1054 - Software Project Management ✓	Elective - I MTIT1052- Mobile Computing ✓	Elective-II MTEC1052 - Image and Video Processing ✓	Elective - I MTNE1051 - Advanced Fluid Mechanics	Elective - I MTEE1051 - Reactive Power Management ✓
	Supplementary	Elective-I MCS1052-Software Engineering ✓	Elective-I MIT1052-Mobile Computing ✓	Elective - I MEC1051-DSP Processors and Architecture ✓ MEC1052-Image and Video Processing ✓	Elective - I MME1051-Gas Dynamics 2	Elective - I MEE1051-Modern Control Theory 2
10-04-2015 Friday	Regular	Elective-II MTCS1063 - Artificial Intelligence ✓	Elective - II MTIT1063 - Digital Image Processing ✓	Elective-II MTEC1061 - Bio- Medical Signal Processing ✓	Elective - II MTNE1061 - Advanced Power Plant Engineering	Elective - II MTEE1061 - Industrial Electronics ✓
	Supplementary	Elective-II MCS1061-Automata and Compiler Design ✓ MCS1063-Artificial Intelligence ✓	Elective-II MIT1063-Digital Image Processing ✓	Elective - II MEC1061-Radar Signal Processing ✓	Elective - II MME1061-Solar Energy ✓	Elective - II MEE1061-Digital controllers for Power Electronic systems 2

NOTE: (i) Any omissions or clashes in this time table may please be informed to the Controller of Examinations immediately.
(ii) Even if government declares holiday on any of the above dates, the examinations shall be conducted as usual.
(iii) For any clarification in respect of the above examinations please contact Controller of Examinations.

Date : 14-03-2015

Copy to: 1.M.TECH. Notice Boards 2. M.TECH. H.O.Ds for n.a.,

DIRECTOR

M.TECH. I Semester Regular/Supplementary Examinations

MTEE101 – MACHINE MODELING AND ANALYSIS

Time: 3 hours

Max. Marks: 60

Answer all the questions
All Questions carry equal Marks

30 MAR 2015

- 1(a) Obtain an expression for the electrical torque of the Kron's primitive machine. Show that no torque is produced by interaction between the flux and current on the same axis. 12 M

OR

- 1(b) What is Kron's primitive machine? How are the various windings of a machine represented by the primitive machine? 12 M

- 2(a) A separately excited d.c. motor is running at steady speed ω_1 with its moment of inertia J_1 and damping co-efficient D . Now an inertia load J_2 is suddenly applied to the motor shaft by means of a clutch. Find the energy loss at the clutch. 12 M

OR

- 2(b) A d.c. shunt motor can be quickly brought to rest by means of plugging, which involves reversing the polarity of the armature supply, the field circuit maintaining the constant field current as before. Obtain an expression for the speed as a function of time. 12 M

- 3(a) A three phase induction motor has the following per phase parameters referred to stator:
Stator resistance = 0.4Ω , Rotor resistance = 0.54Ω , stator and rotor leakage reactance = 2.5Ω each and magnetizing reactance = 40Ω . Find out the parameters of an equivalent 2 – phase induction motor if its per phase turns are same as that of the 3 – phase induction motor. 12 M

OR

- 3(b) In order to ensure power invariance in transforming one set of variables to another, show that the transpose of the transformation matrix should be equal to its inverse. 12 M

- 4(a) Draw the basic circuit model for a 3-phase induction motor for stator as well as rotor and obtain voltage equations in the form of matrices in terms of stator and rotor currents. 12 M

OR

- 4(b) Derive and obtain expressions for flux linkages in the two axis model for a 3-phase induction motor from ϕ_a, ϕ_b, ϕ_c values. 12 M

- 5(a) Derive the steady state voltage equation for the synchronous machine using the stator voltage equations expressed in the arbitrary reference. 12 M

OR

- 5(b) A 3-phase, 50Hz cylindrical rotor synchronous machine has the following parameters:
Self inductance for phase 'a' = 2.45 mH .
Armature leakage inductance = 0.55 mH .
For this machine, calculate the mutual inductance between armature phases and its synchronous reactance. 12 M

M.TECH. I Semester Regular/~~Supplementary~~ Examinations

MTME101 – MATHEMATICAL METHODS IN ENGINEERING

Time: 3 hours

Max. Marks: 60

Answer all the questions
All Questions carry equal Marks

30 MAR 2015

- 1(a) Find the inverse of the matrix $A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 2 \end{bmatrix}$ using Gauss-Jordan elimination method. 4M

- (b) Solve the following system by Gauss-Seidel iteration method. 8M
 $20x + y - 2z = 17$; $3x + 20y - z = -18$; $2x - 3y + 20z = 25$

OR

- 1(c) Solve the following system of equations by Gauss-Seidel iteration method. 12M
 $10x + 2y + z = 9$; $2x + 20y - 2z = -44$; $-2x + 3y + 10z = 22$

- 2(a) Find the smallest eigenvalue in magnitude of the matrix $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ using four iterations of the inverse power method. 6M

- (b) Find the eigenvalues and the corresponding eigenvectors of the matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$. 6M

OR

- 2(c) Verify Cayley Hamilton theorem and find the inverse of the matrix $\begin{bmatrix} 1 & 3 & 7 \\ 1 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$. 12M

- 3(a) If $f(x) = e^{ax}$, then show that $\Delta^n e^{ax} = (e^{ah} - 1)^n e^{ax}$, where h is the step size. 4M

- (b) The velocity v (km/min) of a moped which starts from rest, is given at fixed intervals of time t (min) as follows 8M

t	2	4	6	8	10	12	14	16	18	20
v	10	18	25	29	32	20	11	5	2	0

OR

- 3(c) Evaluate $\int_0^{\pi/2} e^{\sin x} dx$ taking $h = \pi/6$. 6M

- (d) Using Romberg's integration method, find the value of $\int_1^{1.8} y(x) dx$ starting with trapezoidal rule, for the tabular values. 6M

X	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8
y(x)	1.543	1.669	1.811	1.971	2.151	2.352	2.577	2.828	3.107

- 4(a) A boundary value problem is defined by $y'' + y + 1 = 0$, $0 \leq x \leq 1$ where $y(0) = 0$ and $y(1) = 0$ with $h = 0.5$, use the finite-difference method to determine the value of $y(0.5)$. 12M

OR

- 4(b) Using the method of separation of variables solve $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$, where $u(x, 0) = 6e^{-3x}$ 12M

- 5(a) Solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$; $u(0, t) = u(1, t) = 0$; $u(x, 0) = \sin \pi x$, $0 \leq x \leq 1$ using Schmidt method. 12M

OR

- 5(b) Find the solution of the parabolic equation $u_{xx} = 2u_t$ when $u(0, t) = u(4, t) = 0$ and $u(x, 0) = x(4 - x)$, taking $h = 1$. Find the values upto $t = 4$. 12M

M.TECH. I Semester Regular/~~Supplementary~~ Examinations

MTEC101 – ADVANCED DIGITAL SIGNAL PROCESSING

Time: 3 hours

Max. Marks: 60

Answer all the questions
All Questions carry equal Marks

30 MAR 2015

- 1(a) Explain about the necessity of sampling rate conversion. 6 M
- (b) With a neat diagram explain about the implementation of sampling rate conversion by FIR Filter. 6 M

OR

- 1(c) Define Nobel identities and their applications with different examples. 6 M
- (d) If the Z transform of a sequence $x(n)$ is $X(z)$ then what is the Z transform of a sequence up sampled by a factor L. 6 M

- 2(a) Write about Estimation of power spectral density from finite data. 6 M
- (b) What are the difficulties in non-parametric methods for measuring the spectral density? 6 M

OR

- 2(c) Compare the performance measures of the nonparametric methods of power spectrum estimates. 6 M
- (d) Explain about the computational requirements of nonparametric power spectrum estimates. 6 M

- 3(a) Discuss about the Burg method for AR parameter estimation. 6 M
- (b) Mention the limitations of Burg method. 6 M

OR

- 3(c) With a neat diagram explain about the Non-linear least squares method. 12 M

- 4(a) Explain how Filter bank structure is used in the interpretation of periodogram. 12 M

OR

- 4(b) Define cross correlation and cross spectrum .Relate the output cross spectrum in terms of the input cross spectrum while filtering a random process. 12 M

- 5(a) Explain the process of finding the FIR Wiener filter co-efficient that minimize the mean square error and derive the necessary equations. 12 M

OR

- 5(b) Give the properties of linear estimators and the Cramer-rao bound. 12 M

M.TECH. I Semester Regular/~~Supplementary~~ Examinations

MTCS101 – ADVANCED DATA STRUCTURES

30 MAR 2015

Time: 3 hours

Max. Marks: 60

Answer all the questions
All Questions carry equal Marks

- 1(a) Explain the operations of Queue ADT with an example. 6 M
- (b) Write an algorithm for Forest traversals 6 M
- OR**
- 1(c) Write an algorithm to insert an element in front of a circular linked list. 6 M
- (d) Explain the deletion operation of Binary search tree with an example. 6 M
- 2(a) Explain the Construction of an AVL for the following elements. 12 M
20,10,40,3,2,7,60,1,80,60,15,22
- OR**
- 2(b) Write an algorithm to delete an element from the binary search tree. 6 M
- (c) Explain the splaying operations with an example. 6 M
- 3(a) Sort the following list elements by using Quick Sort 65, 80, 90, 70, 55, 35, 20, 45 6 M
- (b) Explain melding operation of Leftist tree. 6 M
- OR**
- 3(c) Sort the following list of elements by using Insertion sort 6 M
26, 5,77,1,61,11,59,15,48,19
- (d) Explain how to delete Min element from the Binomial Heap. 6 M
- 4(a) Describe the prim's algorithm for finding shortest paths in minimum cost spanning tree. 6 M
- (b) Solve the 0/1 knapsack problem for $n = 3$, $(P_1, P_2, P_3) = (1, 3, 4)$, $(w_1, w_2, w_3) = (2, 4, 8)$, and $m = 10$. 6 M
- OR**
- 4(c) Explain minimum cost spanning tree using prim's and kruskal's algorithm with an example. 12M
- 5(a) Write LCBB algorithm to find minimum cost answer node. 6 M
- (b) Draw and explain the tree organization of the 4-queen solution space. 6 M
- OR**
- 5(c) Draw a portion of state space tree for solving sum of subsets problem for $n=5, m=21$ and $(w_1, w_2, w_3, w_4, w_5) = (5, 6, 10, 11, 16)$. 6 M
- (d) Differentiate between backtracking and branch and bound method. 6 M

MTIT101 – ADVANCED PROBLEM SOLVING

Time: 3 hours

Max. Marks: 60

Answer all the questions
All Questions carry equal Marks

30 MAR 2015

- 1(a) Why do we use asymptotic notations in the study of algorithms? 5 M
- (b) Briefly describe the commonly used asymptotic notations? 7M

OR

- 1(c) List out any two applications of linked list and two advantages of doubly linked lists over singly linked list? 6 M
- (d) Convert the following postfix expression into its corresponding infix and prefix expression. ABCDE/*-F/G++ 6 M

- 2(a) Use a stack to evaluate the following postfix arithmetic expression. Show the changing status of the stack in tabular form: 8 M
X Y Z ^ * + A B / C + -for X=1, Y=5, Z=2, P=3, A=15, B=3, and C=8
- (b) Differentiate between Stack and Queue. How are they related to LIFO and FIFO concept? 4 M

OR

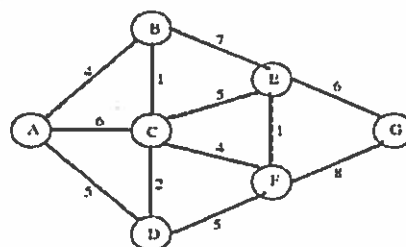
- 2(c) A priority queue is implemented as a Max-heap. Initially it has 5 elements. The level order traversal of the heap is given below: 10, 8, 5, 3, 2 Two new elements '1' and '7' are inserted in the heap in that order. What is the level - order traversal of the heap after the insertion of these elements? 12M
- 3(a) Describe any two Hash functions using suitable examples. 6 M
- (b) With an example explain insertion sort algorithm in detail? 6 M

OR

- 3(c) Define Linear and Quadratic Probing? 6 M
- (d) Compare and contrast between any four sorting techniques? 6 M
- 4(a) Find the binary tree whose inorder and preorder traversals is given below 12M
inorder = g d h b e i a f j c preorder = a b d g h e i c f j
- (b) Write the disadvantages of linear representation of binary tree? 12M

OR

- 4(c) For the following graph calculate the shortest path from start vertex 'A' using theDijkstra's Algorithm 12M



- 5(a) Write and explain Knuth-Morris-Pratt Algorithm and discuss its computational complexity? 12M

OR

- 5(b) Draw BSTs of height 2, 3 and 4 on the set of keys { 10,4,5,16,1,17,21} 6 M
- (c) Define B-Tree. Show all legal B-Trees of minimum degree 2 that represent {1,2,3,4,5}. 6 M

M.TECH. I Semester Regular/~~Supplementary~~ Examinations

MTCS102 – ADVANCED DATABASE MANAGEMENT SYSTEMS

Time: 3 hours

Max. Marks: 60

Answer all the questions
All Questions carry equal Marks

1 APR 2015

- 1(a) Discuss about query optimization briefly. 6M
(b) Explain various stages in query decomposition. 6M
- OR**
- 1(c) What are merits and demerits of lock-based concurrency control? Describe how timestamp based protocol addresses these demerits. 12M
- 2(a) Describe the necessity of parallel databases. List its features. 6M
(b) Explain two-phase commit protocol. 6M
- OR**
- 2(c) Discuss distributed transaction model for DDBMS with a diagram 6M
(d) Explain why 3-phase commit protocol is non-blocking when compared to 2-phase commit protocol. 6M
- 3(a) What is a persistent programming language? How does it differ from an OODBMS? 6M
(b) How OODBMS addresses the disadvantages of RDBMS? 6M
- OR**
- 3(c) List the advanced database applications for which existing RDBMSs are proven to be inadequate. 6M
(d) Write a short note on Object Definition Language. Give an example to illustrate the discussion. 6M
- 4(a) What is data mining? How is it different from data retrieval through query languages? 6M
(b) Compare OLTP systems and Data Warehousing. 6M
- OR**
- 4(c) Briefly explain data warehousing tools and technologies. 6M
(d) Briefly explain information flows of a data warehouse. 6M
- 5(a) What is a knowledge base? What is its significance? 6M
(b) List and compare various multimedia data structures 6M
- OR**
- 5(c) Write a short note on spatial databases 6M
(d) Compare and contrast proof-theoretic and model-theoretic interpretation of rules. 6M

M.TECH. I Semester Regular/Supplementary Examinations

MTIT102 – SOFTWARE ARCHITECTURE

Time: 3 hours

Max. Marks: 60

Answer all the questions
All Questions carry equal Marks

19 APR 2015

- 1(a) What is Software architecture? Discuss in detail about the issues in architecture? 6 M
- (b) Write a short note on data repositories? 6 M
- OR**
- 1(c) Discuss in detail about various Architecture Quality Attributes Used for Evaluation & Analysis? 12M
- 2(a) With a neat diagram explain the internal structure of interactive update process? 6 M
- (b) Describe in detail about Virtual layers in detail? 6 M
- OR**
- 2(c) Discuss briefly about traditional compiler? 6 M
- (d) What are the tools that could interact with shared repository? Explain. 6 M
- 3(a) Explain the steps involved in formalizing the architectural style in detail? 12M
- OR**
- 3(b) Discuss about the notational linguistic issues in software architectural design? 12M
- 4(a) Describe various requirements for architectural description language? 12M
- OR**
- 4(b) Explain the ideal properties desirable from architectural description language? 12M
- 5(a) Describe how architecture interconnection takes place? Give Examples 12M
- OR**
- 5(b) What are the various component types supported in Unicon? Explain 12M

M.TECH. I Semester Regular/~~Supplementary~~ Examinations

MTEC102 – DIGITAL SIGNAL PROCESSORS

Time: 3 hours

Max. Marks: 60

Answer all the questions
All Questions carry equal Marks

1 APR 2015

- 1(a) Explain the issues to be considered in designing and implementing a DSP system, with the help of a neat block diagram. 6 M
- (b) Describe A/D and D/A conversion errors. 6 M
- OR**
- 1(c) Explain about the Hardware Architecture for DSP Processors and its interfacing. 8 M
- (d) What are the applications of on-chip timer in P-DSPs? 4 M
- 2(a) Explain the general purpose registers in TMS320C6X processor and also explain how the 40-bit is read and written from the registers. 6 M
- (b) Write short notes about arithmetic logic unit and accumulator. 6 M
- OR**
- 2(c) What is meant by circular addressing mode? Write pointer updating algorithm for the Circular addressing mode and show different cases that encounter during the updating process of the pointer. 6 M
- (d) Explain the Architectural features of TMS320C6X floating point processor. 6 M
- 3(a) Obtain the impulse response of the analog filter whose transfer function is given by $H(s) = \frac{1}{(s-0.75)}$. If the sampling rate used is 10,000 samples/sec, what is the impulse response of the equivalent digital filter obtained using the impulse invariant method. 12 M
- OR**
- 3(b) Write a MATLAB program to multiply two 3 x 3 matrices. 6 M
- (c) What are the finite word length effects in FIR and IIR filters? 6 M
- 4(a) Write the subroutine for bit reversed address generation. Explain the same. 6 M
- (b) How many add/subtract and multiply operations are needed to compute the butterfly Structure? 6 M
- OR**
- 4(c) Write a MATLAB program for to calculate the DFT of a given sequence using FFT, $x(n) = [2 \ 2 \ 2 \ 2 \ 1 \ 1 \ 1 \ 1]$; Also plot the magnitude and phase plot. 12 M
- 5(a) Design a data memory system with address range 000800h – 000fffh for a c5416 processor using 2kx8 SRAM memory chips. 6 M
- (b) Draw the timing diagram for transmit and receive in synchronous serial interface. 6 M
- OR**
- 5(c) How many address lines are required to access all location of an 16K x 16 SRAM. 6 M
- (d) Write an instruction sequence to reset and disable the transmitter and receiver for the McBSP. 6 M

MTME102 – ADVANCED THERMODYNAMICS

7 APR 2015

Time: 3 hours

Max. Marks: 60

Answer all the questions
All Questions carry equal Marks

- 1(a) Is it possible to attain zero temperature on the absolute thermodynamic temperature scale? Why? 6M
- (b) How would you explain concept of entropy to a lay person? 6M
- OR**
- 1(c) Derive the expression for change in entropy of universe. 6M
- (d) Write a short note on third law of thermodynamics. 6M
- 2(a) What is Joule Kelvin Coefficient? Why is it zero for an ideal gas? 6M
- (b) What is Keenan function? Explain 6M
- OR**
- 2(c) Differentiate between availability function and Gibbs free energy function. 6 M
- (d) Prove that available energy decreases when heat transfers through finite temperature difference. 6 M
- 3(a) What is a compressible fluid? What are the basic laws in compressible flow? 6M
- (b) What is stagnation state? What do you mean by stagnation properties? 6M
- OR**
- 3(c) Comprehend the Mach Number and Mach Angle 6M
- (d) Air is accelerated isentropic ally from 100 m/s to 400 m/s in a nozzle .If the temperature at the initial state is 400 K and Mach Number is 1.5 , Determine i) Initial Mach Number ii) Final Mach Number. 6M
- 4(a) A Gaseous mixture consists of 1 kg of oxygen and 2 Kg of nitrogen at a pressure of 150 KPa and a temperature of 20° C. Determine the changes in internal energy, enthalpy and entropy of the mixture when the mixture is heated to a temperature of 100°C at (a) constant volume and (b) constant pressure. 12M
- OR**
- 4(b) Two spheres each 2.5 m in diameter are connected to each other by a pipe with a valve. One sphere contains 16 kg of air and other 8 kg of air when the valve is closed. The temperature of the air in both spheres is 25 °C. The valve is opened and the whole system is allowed to come to equilibrium conditions. Assuming there is no loss or gain of energy determines the pressure in the spheres when the system attains equilibrium. Neglect the volume of the pipe. 12 M
- 5(a) List out the properties required for an ideal binary fluid used in high temperature limit. 6 M
- (b) Explain the concept of binary vapour cycle in brief. 6M
- OR**
- 5(c) What is cogeneration plant? What are the thermodynamic advantages of such a plant? 6M
- (d) What is the biggest loss in a steam power plant? How can these losses be reduced? 6M

M.TECH. I Semester Regular/Supplementary Examinations

MTEE102 – POWER CONVERTERS

11 APR 2015

Time: 3 hours

Max. Marks: 60

Answer all the questions
All Questions carry equal Marks

- 1(a) Explain single- phase half wave rectifier for RL load with suitable voltage and current wave forms. 6 M
- (b) Derive the expressions for average output voltage, RMS load current. 6 M
- OR**
- 1(c) Explain the operation of PWM rectifier in bridge connection with the help of neat diagram? 6 M
- (d) A single phase fully controlled bridge circuit is used for obtaining a regulated d.c output voltage. The RMS value of the a.c input voltage is 230 V, and the firing angle is maintained at $\pi/3$, so that the load current is 6A. Calculate the d.c output voltage. 6 M
- 2(a) Derive the expression for the following performance factors of 3 – phase fully controlled bridge converter: 12 M
- i) Power factor ii) Harmonic factor iii) Distortion factor.
- OR**
- 2(b) What are the harmonic standards and recommended practices to limit the magnitudes of harmonic currents and harmonic voltage distortion? 12 M
- 3(a) A single phase full wave voltage controller is used for controlling the power from 230 V, 50Hz. Source into a load circuit consisting of $R=8\ \Omega$ and inductor of reactance $\omega L=6\ \Omega$, calculate 12 M
- (a) Control range of firing angle
- (b) The maximum value of rms load current
- (c) Maximum value of average (dc) and rms thyristor currents.
- OR**
- 3(b) Draw the circuit of single phase full wave ac voltage controller with R-L load? 6 M
- (c) Derive expressions for rms output voltage V_o , harmonic output voltage V_o , harmonic output current, functional frequency output voltage and current. 6 M
- 4(a) What is pulse – width modulation? List the various PWM techniques. How do these two differ from each other? 12 M
- OR**
- 4(b) What is the need for controlling the output at the output terminals of an inverter? Discuss briefly and compare the various methods employed for the control of output voltage of inverters. 12 M
- 5(a) A single – phase flying – capacitor multi level inverter has $m=5$. Find the number of capacitors, the peak voltage and current ratings of diodes and switching devices if $V_{dc}=5\text{kv}$. 12 M
- OR**
- 5(b) A single – phase diode – clamped inverter has $m=5$. Find the peak voltage and current ratings of diodes and switching devices, if $V_{dc}=5\text{ kv}$ and $i_o=50 \sin(\theta - \pi/3)$. 12 M

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L.B.Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech I Semester Regular/~~Supplementary~~ Examinations

MTCS103-DATA MINING
(Computer Science and Engineering)

4 APR 2015

Time : 3 hours

Max.Marks :60

Answer all the questions
All Questions carry equal Marks

- 1(a) Write short notes on Feature Subset Selection? [6M]
(b) Discuss the various types of Datasets? [6M]

(OR)

- (c) Explain briefly about various Summary Statistics Techniques? [12M]
2(a) Explain the Bayesian classification with an illustration. [12M]

(OR)

- (b) Explain the method for expressing the Attribute Test Condition? [12M]
3(a) Explain the classification criteria of Frequent Pattern Mining? [6M]
(b) Generate the frequent item sets using candidate generation for the following database. [6M]

TID	List if items
T100	I1, I3, I4
T200	I2, I3,I5
T300	I1,I2, I3,I5
T400	I2, I5

(OR)

- (c) Differentiate Fp-Growth and Apriori algorithm for generating itemsets. Explain the pros and cons of these algorithms. [12M]
4(a) Explain the different types of clusters? [6M]
(b) Explain Bisecting K-Means algorithm briefly? [6M]

(OR)

- (c) Define Cluster evaluation? What are the important issues of Cluster evaluation? What are the types of evaluation measures? [12M]
5(a) Write short note on Enterprise search? [6M]
(b) What are the various characteristics of Search Engines? [6M]

(OR)

- (c) Explain the two major steps of HITS algorithm? What are the problems occurred in HITS algorithm? [12M]

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M.Tech I Semester Regular/~~Supplementary~~ Examinations

MTIT103 – SOFTWARE PROCESS MANAGEMENT
(Software Engineering)(IT)

4 APR 2015

Time : 3 hours

Max.Marks :60

Answer all the questions
All Questions carry equal Marks

- 1(a) Explain about the people in the optimizing process. [6M]
(b) Distinguish between managed process and optimizing process. [6M]

(OR)

- (c) Which process model has the cost of gathering data? Explain. [6M]
(d) List the merits and demerits of the repeatable process maturity level. [6M]
2(a) Describe Wideband Delphi technique. [6M]
(b) How to establish a project management system? Explain. [6M]

(OR)

- (c) Explain about software quality management. [6M]
(d) What is Change Control Board? Explain in detail. [6M]
3(a) Give examples of some major standards. [6M]
(b) Describe the requirements phase of the SCM. [6M]

(OR)

- (c) Summarize the concepts of software inspections. [12M]
4(a) What are software measures? Explain in detail. [12M]

(OR)

- (b) Explain in detail about establishing a software quality program. [12M]
5(a) Discuss about process changes for defect prevention. [6M]
(b) What is the need for software automation? Explain. [6M]

(OR)

- (c) Briefly explain defect prevention to optimize the software process. [12M]

MTEC103- Linear Algebra for Signal Processing

(Systems and Signal Processing)(ECE)

Time : 3 hours

Max.Marks :60

Answer all the questions
All Questions carry equal Marks

4 APR 2015

- 1(a) What is the algebraic signal processing theory? [6M]
(b) Show that the product of two diagonal matrices is a diagonal matrix. [6M]

(OR)

- (c) Find the adjoint and inverse of [6M]

$$A = \begin{pmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{pmatrix}$$

- (d) find matrix transform and prove $(AB)^T = B^T A^T$ [6M]

- 2(a) Let 0 represent OFF and 1 represent ON and [6M]

$$A = \begin{pmatrix} ON & ON & OFF \\ OFF & ON & OFF \\ OFF & ON & ON \end{pmatrix}$$

Find the ON/OFF matrix B so that A+B is a matrix with each entry ON.

- (b) Without computing A. find bases for the four fundamental subspaces. [6M]

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 6 & 1 & 0 \\ 9 & 8 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{pmatrix}$$

(OR)

- (c) If a, b, c are given with $a \neq 0$, how would you choose d so that $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ has rank one?
Find a basis for the row space and null space. [12M]

- 3(a) Put bases for the subspaces V and W into the columns of matrices V and W. Explain why the test for orthogonal subspaces can be written $V^T W = \text{zero matrix}$. This matches $u^T w = 0$ for orthogonal vectors. [12M]

(OR)

- (b) Derive Toeplitz matrices and explain Shift invariant systems using noncausal system. [12M]

- 4(a) Find the four Eigen values and Eigen vectors of given matrices: [12M]

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

(OR)

- (b) A normal matrix has $A^T A = A A^T$; it has orthogonal Eigen vectors. Why is every skew-symmetric matrix normal? Why is every orthogonal matrix normal?
When is $\begin{pmatrix} a & 1 \\ -1 & d \end{pmatrix}$ normal? [12M]
- 5(a) Discuss various linear transformation mapping R^n and R^m . [6M]
(b) If $L: R^2 \rightarrow R^2$ be defined by $L(x, y) = (x+y, x-y)$. Determine whether mL is one-to-one? [6M]
- (OR)**
- (c) Explain Linear Algebra application in [12M]
a) Transformations
b) Statistics and Probability

MTEE103-CONTROL OF MOTOR DRIVES - I

(Power Electronics and Drives)(EEE)

Time : 3 hours

Max.Marks :60

Answer all the questions
All Questions carry equal Marks

4 APR 2015

- 1(a) Explain in detail the operation of a full converter, feeding a separately excited d.c. motor with reference to voltage and current waveforms assuming that the motor current is continuous. [12M]
(OR)
- (b) Three phase converters are better than single phase converters when harmonics are considered. Will this advantage be of any use in motor drives? [12M]
- 2(a) For the class A chopper circuit derive an expression for the output power with resistive load. Calculate the per unit value of this for [12M]
(i) $\gamma \left(\frac{1}{4}\right)$ (ii) $\gamma \left(\frac{3}{4}\right)$.
(OR)
- (b) In a class A chopper circuit with battery V_s , with resistive load and duty cycle γ , sketch waveforms of the voltages across the switch and the diode. Derive expressions for the average values of these. [12M]
- 3(a) For variable frequency control of induction motor, explain the following points:
(i) For speeds below base speed (v/f) ratio is maintained constant, why?
(ii) For speeds above base speed, the terminal voltage is maintained constant, why? [12M]
(OR)
- (b) Write merits and demerits of Voltage source fed induction motor control and current source fed inverter control of induction motor. [12M]
- 4(a) Why the rotor resistance control is preferred in low power crane drives? How does the rotor resistance control help during counter – torque braking. [12M]
(OR)
- (b) A 3 – phase, 400V, 6-pole, 50Hz, delta connected, slip ring induction motor has rotor resistance of 0.3Ω and leakage reactance of 1Ω per phase referred to stator. When driving a fan load it runs at full load at 4% slip. What resistance must be inserted in the rotor circuit to obtain a speed of 900 rpm. Neglect stator impedance and magnetizing branch. Stator to rotor turns ratio is 2.2. [12M]
- 5(a) Develop a procedure to synthesize the torque and flux component current controllers for a direct vector controlled induction motor drive. What are the salient differences between these? [12M]
(OR)
- (b) A flux – weakening controller is least parameter – sensitive in the stator and rotor flux – linkages – based vector controllers. Comment on the veracity of this statement. [12M]

MTME103-ADVANCED HEAT AND MASS TRANSFER

(Thermal Engineering)(ME)

Time : 3 hours

Max.Marks :60

Answer all the questions
All Questions carry equal Marks

4 APR 2015

- 1(a) A copper pipe carries process steam at 100°C and is 25mm OD and 22mm ID through a room which is at 30°C . The cost of heating by the steam is 5 paise per 4×10^3 kJ. Standard 85 % magnesia insulation is available having ID of 25 mm and OD of 40 mm and costs Rs.4 per metre length of pipe. How long must the pipe be in operation to recover the insulation cost. The resistance of the fluid film is roughly constant and equals 0.05 m-K/W [6M]

- (b) Compare and contrast the efficiency of fin and effectiveness of fin. [6M]

(OR)

- (c) A furnace wall is made of 75 mm thick fire clay brick and 64 mm thick mild steel plate. The inside surface of brick is at 647°C and outside air temperature is 27°C .

Estimate:

- a) The heat loss per m^2 area of the furnace wall and outside surface temperature of the steel plate.
b) If 18 steel bolts each of 19 mm diameter are used for fixing the steel plate on the brick wall per m^2 area of the wall, calculate the percentage increase in the heat flux? [12M]

- 2(a) A large slab of aluminium ($\alpha = 8.4 \times 10^{-3} \text{ m}^2/\text{hr}$, $k = 215 \text{ W/m K}$) is at a uniform temperature of 200°C . Suddenly its surface temperature is lowered to 70°C . How much amount of heat will escape from the slab per unit surface area when the temperature at a depth of 40mm has dropped to 120°C ? [12M]

(OR)

- (b) A house hold iron has a surface area of 0.05 m^2 and is made of stainless steel with total weight of 1.4 kg and 500 W capacity. The heat transfer coefficient on the surface of the iron its surroundings is $17 \text{ W/m}^2\text{-}^{\circ}\text{C}$ when surrounding air temperature is 30°C . How long after being turned on will the iron reach 110°C if it is originally at surrounding temperature?

Take ρ (iron steel) $= 7820 \text{ kg/m}^3$. K (iron steel) $= 17.3 \text{ W/m-}^{\circ}\text{C}$. C_p (iron steel) $= 460.8 \text{ J/kg-}^{\circ}\text{C}$. [12M]

- 3(a) Define the bulk temperature. How is it used ? [6M]

- (b) Explain significance of Grashoff number in internal flows? [6M]

(OR)

- (c) Air flows over a flat plate at a velocity of 3 m/sec and ambient conditions are, pressure is 760 mm of Hg and temperature is 15°C . The plate is maintained at 85°C . If the length of the plate is 100 cm along the flow of air, find the heat lost by 50 cm of the plate which is measured from the trailing edge. Plate width is 50 cm.

Properties of air at mean temperature of 50°C

$\rho = 1.093 \text{ kg/m}^3$, $c_p = 1007 \text{ J/kg-K}$

$K = 0.0278 \text{ W/m-K}$, $\nu = 17.95 \times 10^{-6} \text{ m}^2/\text{sec}$, $Pr = 0.71$

[12M]

- 4(a) What are the types of boiling processes? [6M]

- (b) What are the types of condensation processes? [6M]

(OR)

- (c) Air free saturated steam at 90°C condenses on the outer surface of a 1.5m long 2.5cm OD vertical tube maintained at a uniform temperature of 70°C . Assume film condensation.

Estimate

1. Heat transfer coefficient at the end of plate ($h_{x=L}$)
2. Mean heat transfer coefficient
3. Thickness of the film at the end of tube (δ_L)
4. Mass of condensate (m)

Reynolds number at the bottom (Re)

[12M]

- 5(a) How is the diffusion coefficient defined? [6M]

- (b) Explain the term "mass transfer" and what are the modes of mass transfer? [6M]

(OR)

- (c) A vessel contain a binary mixture of O_2 and N_2 with partial pressure in the ratio of 0.21 and 0.79 at 15°C . The total pressure of the mixture is 1.1 bar. Calculate the following

1. Molar concentration
2. Mass densities
3. Mass fractions
4. Molar fractions

[12M]

MTEE104-COMPUTATIONAL MATHEMATICS

(Power Electronics and Drives)(EEE)

Time : 3 hours

Max.Marks :60

Answer all the questions
All Questions carry equal Marks

L 6 APR 2015

- 1(a) Compute the QR decomposition of $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ [12M]

(OR)

- (b) Solve the system of linear equations by least square method. [12M]
 $x + y + z = 6, 2x + 3y - 2z = 2, 5x + y + 2z = 13.$

- 2(a) Find $Z(\sin an)$ and $Z(\cos an)$, $n \geq 0$. [6M]

- (b) Find the inverse Z transform of $\frac{z^3}{(z+1)(z-1)^2}$ using partial fractions. [6M]

(OR)

- (c) Evaluate $Z^{-1} \left[\frac{z^2}{((z-a)(z-b))} \right]$, $(a \neq b)$ by using convolution theorem [12M]

- 3(a) Obtain the Fourier series expansion of $f(x) = e^{-x}$ in $(0, 2\pi)$. [12M]

(OR)

- (b) Represent the function $f(x) = \begin{cases} -x^2, & -\pi \leq x \leq 0 \\ x^2, & 0 \leq x \leq \pi \end{cases}$ as a Fourier series. [12M]

- 4(a) Distinguish between linear and circular convolution of two sequences.[6M]

- (b) What do you understand by periodic function? [6M]

(OR)

- (c) Find the IDFT of the sequence

$$x(n) = \{4, 1 - j2.414, 0, 1 - j0.414, 0, 1 + j0.414, 0, 1 + j2.414\} \text{ Using DIF algorithm.}$$

[12M]

- 5(a) Solve the Euler's equation for the function $\int_{x_0}^{x_1} (x + y^1). y^1. dx$. [12M]

(OR)

- (b) Find the curves on which the functional $\int_0^{\frac{\pi}{2}} ((y^1)^2 - y^2 + 2xy). dx$ with $y(0) = 0$ and $y(\frac{\pi}{2}) = 0$ can be extremised. [12M]

MTME104-ADVANCED IC ENGINES

(Thermal Engineering)(ME)

Time : 3 hours

Max.Marks :60

Answer all the questions
All Questions carry equal Marks

6 APR 2015

- 1(a) Derive the necessary expression for correction factors for power and volumetric efficiency. [6M]
(b) What is MBT? Explain its significance. [6M]
- (OR)**
- (c) List out the important differences between the design and operating characteristics of two stroke and four stroke engine. [12M]
- 2(a) Explain the process of combustion in dual fuel operation. [6M]
(b) What are the advantages and disadvantages of HCCI engines. [6M]
- (OR)**
- (c) Explain the advantages of variable compression ratio engines. [6M]
(d) Compare a rotary Wankel engine with reciprocating engine. [6M]
- 3(a) Explain why SI engine torque varies with spark timing?
What is the best spark timing? Explain how it varies with speed and load.[12M]
- (OR)**
- (b) Discuss in detail the differences between abnormal combustion in SI and CI engines. [12M]
- 4(a) Explain the possibilities of using dual fuel systems in IC engines. [6M]
(b) Can we use solid fuels for IC engines? If yes, how? If no, why? [6M]
- (OR)**
- (c) In a test on two stroke oil engine, the following results were obtained: speed = 350 rev/min; Net brake load = 600 N; Mean effective pressure = 2.66 bar; Fuel consumption = 3.2 kg/h; cooling water used = 495 kg/h; Temperatures of jacket water at inlet and outlet = 30°C and 50°C; Exhaust gases per kg of fuel = 32 kg; Temperature of exhaust gases = 432°C; specific heat of exhaust gases = 1.005 kJ/kg K; Inlet air temperature = 32°C. Draw up a heat balance for the engine if its cylinder diameter = 205 mm and stroke = 275 mm; brake drum diameter = 1.0 m; calorific value of fuel = 40870 kJ/kg. [12M]
- 5(a) Explain in detail the following mechanisms of HC formation (i) the absorption and desorption in engine oil (ii) Flame quenching at the walls. [12M]
- (OR)**
- (b) Write sort note on (i) stages of soot formation (ii) EGR (iii) Measurement of emissions. [12M]

MTEC104 SOFT COMPUTING TECHNIQUES

(Systems and Signal Processing)(ECE)

6 APR 2015

Time : 3 hours

Max.Marks :60

Answer all the questions
All Questions carry equal Marks

- 1(a) Show that the McCulloch-Pitts formal model of a neuron may be approximated by a sigmoid neuron. [7M]
(b) Explain the historical development of artificial neural networks? [5M]
(OR)
(c) Explain the structure and working of a neuron with help of suitable diagrams [6M]
(d) Explain Different Network connections? [6M]
- 2(a) Using the perceptron learning rule, find the weights required to perform the following classification. Vectors (1 1 1 1), (-1 1 -1 -1) and (1 -1 -1 1) are members of class (having target value 1) and vectors (1 1 1 .7 1) and (1 -1 -1 1) are not members of class (having target value -1). Use learning rate of 1 and starting weights zero. [8M]
(b) Define and plot the identity function, binary step function, binary sigmoidal function and bi-polar sigmoidal function, used in neural networks. [4M]
(OR)
(c) Discuss the Convergence issues in the back propagation algorithms. [6M]
(d) Using Hebb rule of discrete BAM, find the weight matrix to store the following input-output pattern pairs : [6M]
 $s_1 = (1, 1, 0,),$ $t_1 = (1, 0)$
 $s_2 = (0, 1, 0,),$ $t_2 = (0, 1)$
- 3(a) Discuss physical significance of energy function used in Hopfield neural network. [6M]
(b) Explain the Characteristics of Associative memory . [6M]
(OR)
(c) Draw the architecture of BAM and give a problem that can be solved with the help of network? [12M]
- 4(a) Write the Advantages of fuzzy logic control over the artificial neural networks [6M]
(b) List different operations and properties of classical sets. [6M]
(OR)
(c) Explain in detail the methods employed for converting fuzzy form into crisp form? [12M]
- 5(a) State the importance of Genetic algorithm. [6M]
(b) How a Boltzman machine works?Mention two applications. [6M]
(OR)
(c) Explain in detail about the convergence of genetic algorithm. [6M]
(d) Compare and contrast 'Boltzmann training' and 'Cauchy training' [6M]

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M.Tech I Semester Regular/~~Supplementary~~ Examinations

MTIT104-OBJECT ORIENTED SOFTWARE ENGINEERING

(Software Engineering)(IT)

Time : 3 hours

Max.Marks :60

Answer all the questions
All Questions carry equal Marks

↑ 6 APR 2015

- 1(a) Briefly discuss about team based organization? [6M]
(b) Give the differences between a client and end user? [6M]

(OR)

- (c) For a library system, Draw a use case diagram that shows which actor performs which use case? Write SRS document? [12M]

- 2(a) Explain various contributing factors for communicating about analysis? [12M]

(OR)

- (b) Consider a traffic light system at a four way cross roads. Assume the simplest algorithm for cycling through the lights. Identify the states of this system and draw the state chart describing them? [12M]

- 3(a) Discuss the activities involved in system design? [6M]
(b) Write a short notes on coupling and cohesion? [6M]

(OR)

- (c) List and explain various groups of design criteria's in detail? [12M]

- 4(a) Write a short notes on main roles involved in reuse? [6M]
(b) Define class implementer, class user and class extender? [6M]

(OR)

- (c) Describe the various model elements used during testing? [12M]

- 5(a) Give an example to explain issue based models and systems? [12M]

(OR)

- (b) What are the two ways to model the configuration item? [6M]
(c) With a neat sketch explain the visualization of organization structure? [6M]

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M.Tech I Semester Regular/~~Supplementary~~ Examinations

MTCS104-ADVANCE COMPUTER NETWORKS

(Computer Science and Engineering)

Time : 3 hours

Max.Marks :60

Answer all the questions
All Questions carry equal Marks

L 6 AFR 2015

- 1(a) Give the brief explanation of Terrestrial microwave system? [7M]
(b) Define Transmission. Explain different types of Transmission. [5M]
- (OR)
- (c) Explain the guided transmission media Twisted pair [8M]
(d) Explain i)Amplitude shift Keying. [4M]
- 2(a) Bring out the difference between OSI and TCP/IP reference models [6M]
(b) Explain the architecture of TCP/IP protocol. [6M]
- (OR)
- (c) Why framing of the bit stream is necessary? Explain different framing techniques with examples. [9M]
(d) If the bit string 0110111110111011111010 is subjected to bit stuffing, what output string will be transmitted? [3M]
- 3(a) Draw and explain ICMP protocols and it's different messages [6M]
(b) Give the format of IP header and explain the significance of each field in detail. [6M]
- (OR)
- (c) Explain Distance vector routing with an example. Also explain count to infinity problem. [12M]
- 4(a) Give the structure of UDP header [7M]
(b) Briefly explain the advantages of SCTP over TCP and UDP. [5M]
- (OR)
- (c) What are the services and features of SCTP? [4M]
(d) Discuss connection establishment and connection release in TCP. [8M]
- 5(a) Explain different types of messages in HTTP transaction in detail. [9M]
(b) Explain inverse domain.. [3M]
- (OR)
- (c) List out functionality of Voice Over IP session and Initiation protocol [8M]
(d) Explain iterative resolving in DNS mapping [4M]

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M.Tech I Semester Regular/~~Supplementary~~ Examinations

MTEC1052-IMAGE AND VIDEO PROCESSING

(Systems and Signal Processing (ECE))

Time : 3 hours

Max.Marks :60

Answer all the questions
All Questions carry equal Marks

8 APR 2015

- 1(a) Explain about basic relationship between pixels? [6M]
(b) Consider the result of a 5×5 uniform averaging filter to a digital image N times. characterise the expected effect [6M]

(OR)

- (c) Describe Image Formation in the eye with Brightness adaption and Discrimination? [6M]
(b) Write a short note on sampling and quantization [6M]
2(a) Explain about different redundancies present in the image.? [6M]
(b) Obtain the frequency response of the following edge detectors
(i) Prewitt operator (ii) sobel operator (iii) laplacian operator [6M]

(OR)

- (c) Discuss about region based image segmentation techniques. Compare threshold region based Techniques [6M]
(d) Explain the methods to overcome the above redundancies [6M]
3(a) Describe about the analog video [6M]
(b) Discuss about the various video formats [6M]

(OR)

- (c) With a neat block diagram explain about the digital video formation [6M]
(d) Differentiate between spatial and temporal resolution [6M]
4(a) Explain about the spatial frequencies [6M]
(b) Explain about the sampling of video in three dimensions [6M]

(OR)

- (c) Give a brief note on Required sampling rates [12M]
5(a) Discuss about general methodologies for two dimensional motion estimation. [12M]

(OR)

- (b) With a neat diagram explain about the Global based Motion Estimation [12M]

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M.Tech I Semester Regular/~~Supplementary~~ Examinations

MTEE1051-REACTIVE POWER MANAGEMENT

(Power Electronics and Drives (EEE))

Time : 3 hours

Max. Marks :60

Answer all the questions
All Questions carry equal Marks

8 APR 2015

- 1(a) Explain the different reactive power bias methods used in compensation. [12M]
(OR)
(b) What is load compensation? Discuss its objectives in power systems. [12M]
- 2(a) What is a static compensator? Explain with neat diagrams about working principles of various types of the static compensators. [12M]
(OR)
(b) Explain the thyristor switched capacitor (TSC) compensation for uncompensated transmission line. [6M]
(c) Explain the switched shunt capacitor compensation for uncompensated transmission line. [6M]
- 3(a) Define reactive power management and explain the mathematical modeling of reactive power dispatching strategy [12M]
(OR)
(b) Write short notes on
(i) Objectives of reactive power coordination. [6M]
(ii) Basic concepts of quality of power supply. [6M]
- 4(a) What are the different load patterns available and also explain the basic methods of load shaping? [12M]
(OR)
(b) Discuss the power tariffs and KVAR based tariffs [12M]
- 5(a) Write short notes on:
(i) Reactive power control requirements for traction systems. [6M]
(ii) Different types of Power tariffs in Demand Side Management (DSM) [6M]
(OR)
(b) What are the characteristics and limitations of capacitors used in domestic applications [12M]

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M.Tech I Semester Regular/~~Supplementary~~ Examinations

MTME1051-ADVANCED FLUID MECHANICS
(Thermal Engineering (ME))

8 APR 2015

Time : 3 hours

Max.Marks :60

Answer all the questions
All Questions carry equal Marks

- 1(a) Derive the integral form of continuity equation starting from the statement of law of conservation of mass for a control volume. [6M]
- (b) Two parallel plates are placed horizontally 10mm apart. The bottom plate is fixed and the top plate is moved at a uniform speed of 0.25 m/s. the fluid between them has a dynamic viscosity μ equal to 1.472 N.s/m². Determine the pressure gradient which corresponds to the condition of zero discharge between the plates and the shearing stress at each plate. [6M]

(OR)

- (c) A tank fitted with a convergent nozzle contains air at a temperature of 20°C. The diameter at the outlet of the nozzle is 25 mm. Assuming adiabatic flow, find the mass rate of flow of air through the nozzle to the atmosphere when the pressure in the tank is
(i) 140 kN/m²(abs) (ii) 300 kN/m²
Take for air: $R=287\text{J/kg } ^\circ\text{K}$ and $\gamma=1.4$ Barometric pressure 100 kN/m². [6M]
- (d) Considering a system and a control volume derive a general equation in differential form for the law of conservation of energy as applied to a fluid flow. [6M]
- 2(a) Assuming second degree velocity distribution in the boundary layer determine using the integral momentum equation, the thickness of boundary layer friction coefficient, displacement and momentum thicknesses. [6M]
- (b) Discuss on displacement thickness of boundary layer. [6M]

(OR)

- (c) Atmospheric air at 25°C flows over a flat plate with a velocity of 2.5 m/s. find the boundary layer thickness at distance of 20cm and 40 cm from the leading edge of the plate. Also find the mass flow that enters the boundary layer between the above two points. The air has a viscosity of 1.8×10^{-5} kg/m.s and a density of 1.20 kg/m³. Assume unit depth in z-direction. [12M]

- 3(a) Explain Mach number, Mach time and Mach cone. Derive energy equation and describe various regions of flow. [6M]
 (b) Determine the distance from the center of the pipe, at which the local velocity is equal to the average velocity for turbulent flow in pipes. [6M]

(OR)

- (c) A smooth pipe of diameter 80 mm and 800 m long carries water at the rate of $0.480 \text{ m}^3/\text{minute}$. Calculate the loss of head, wall shearing stress, center line velocity, velocity and shear stress at 30 mm from pipe wall. Also calculate the thickness of laminar sub-layer. Take kinematic viscosity of water as 0.015 stokes. Take the value of co-efficient of friction 'f' from the relation given as $f = \frac{0.0791}{(R_e)^{1/4}}$, where R_e =Reynolds number [12M]

- 4(a) Discuss briefly the following:
 (i) Mach angle [6M]
 (ii) Mach cone [6M]

(OR)

- (b) Air is released from one tank to another through a convergent-divergent nozzle at the rate of 15 kg/s. the supply tank is at a pressure of 450 kN/m² and temperature 115°C, and the pressure in the receiving tank is 110 kN/m². Estimate (i) The pressure, temperature, and Mach number in the construction, (ii) The required diameter of constriction, (iii) The diameter of the nozzle at the exit for full expansion, and the Mach number. [12M]
 5(a) List out the three governing equations of Fanno flow. [6M]
 (b) Develop an expression for the pressure drop due to friction in a constant area pipe, in terms of initial Mach number, friction factor and L/D ratio of the pipe. [6M]

(OR)

- (c) Derive the expressions for pressure ratio, temperature ratio, velocity ratio, density ratio and stagnation pressure ratio for the Fanno flow of a perfect gas. [12M]

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M.Tech I Semester Regular/~~Supplementary~~ Examinations

MTCS1054-SOFTWARE PROJECT MANAGEMENT
(Computer Science and Engineering)

8 APR 2015

Time : 3 hours

Max.Marks :60

Answer all the questions
All Questions carry equal Marks

- 1(a) Explain the key practices used to improve overall software quality? [6M]
(b) Give the results of conventional software project design reviews? [6M]

(OR)

- (c) What are three levels of process? Mention their attributes? [6M]
(d) Discuss about the three generations of the software development? [6M]
- 2(a) Briefly discuss the outline of software development plan? [6M]
(b) Discuss the primary objectives of inception phase? [6M]

(OR)

- (c) Discuss the measures used to evaluate management set artifacts and requirement artifacts? [12M]
- 3(a) What are the topics included in default content of periodic status assessments? [6M]
(b) Discuss briefly about the workflow of an iteration? [6M]

(OR)

- (c) Write a short note on pragmatic planning process [6M]
(d) Give a neat sketch to explain minor milestones in the life cycle of an iteration? [6M]
- 4(a) Discuss briefly about organization policy outline? [6M]
(b) Discuss briefly about software management team activities? [6M]

(OR)

- (c) Outline the purpose and perspectives of seven core metrics [12M]
- 5(a) Explain all the Software Management principles. [6M]
(b) Show the differences in workflow priorities between small and large projects? [6M]

(OR)

- (c) Give the results of major mile stones in a modern process? [6M]
(d) With a neat diagram show the priorities for tailoring the process framework? [6M]

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M.Tech I Semester Regular/~~Supplementary~~ Examinations

MTIT1052-MOBILE COMPUTING
(Software Engineering,(IT))

8 APR 2015

Time : 3 hours

Max.Marks :60

Answer all the questions
All Questions carry equal Marks

- 1(a) Describe about the multipath propagation [6M]
(b) Explain the data tier in the architecture of the mobile computing [6M]
- (OR)**
- (c) Describe about the direct sequence spread spectrum [6M]
(d) Explain how to find the path loss of radio signals [6M]
- 2(a) Explain the Protocol architecture for signalling of GSM with neat sketch [12M]
- (OR)**
- (b) Explain about the message termination call with neat sketch [6M]
(c) Explain about the security services of GSM [6M]
- 3(a) Explain about the IP – In – IP encapsulation [6M]
(b) Explain why timeout freezing required in case of mobile nodes. What are the modifications made to enforce timeout freezing [6M]
- (OR)**
- (c) Explain about the transaction oriented TCP [6M]
(d) Explain why traditional TCP cannot be used in mobile network [6M]
- 4(a) Explain the frame format of IEEE 802.11 physical frame using FHSS [6M]
(b) What are the applications of MANET [6M]
- (OR)**
- (c) Explain the system architecture of IEEE 802.11 [12M]
- 5(a) Explain the components of a Bluetooth packet at baseband layer [6M]
(b) Explain the basic transaction of WTP class 2 without user acknowledgement [6M]
- (OR)**
- (c) Explain how many types of links does the Bluetooth offers [6M]
(d) Explain the session establishment in WSP/B and also explain the features of session suspension and session resume of WSP/B [6M]

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M.Tech I Semester Regular/~~Supplementary~~ Examinations

MTCS1063-ARTIFICIAL INTELLIGENCE
(Computer Science and Engineering)

10 APR 2015

Time : 3 hours

Max.Marks :60

Answer all the questions
All Questions carry equal Marks

- 1(a) Consider the 4-color problem that is You have to color a planar graph with four colors such that adjacent vertices should not have the color. Give the initial state, goal test, successor function and cost function for the above problem. [12M]

(OR)

- (b) “ Best First search combines the advantage of both depth first search and breadth first search”, Justify it and also give an example. [8M]
(c) What do you mean by PEAS of agents give four examples of agents? [4M]
- 2(a) Write about the following approaches to knowledge representation.
(i) Simple relational knowledge (ii) Inheritable knowledge [6M]
(b) Differentiate weak slot and strong slot. [6M]

(OR)

- (c) What are differences of procedural and declarative knowledge? [6M]
(b) Discuss about scripts using suitable examples? [6M]
- 3(a) Write about Bayes theorem. [8M]
(b) Write about applications of fuzzy set theory [4M]

(OR)

- (c) How we represent knowledge in an uncertain domain. [6M]
(d) What do you mean by measure of belief and hypothesis what is the relation between them explain? [6M]
- 4(a) Explain the following
(i) Realizable learning problem
(ii) Relax and non relax problems
(iii) Genetic algorithms [12M]

(OR)

- (b) What is continues planning? Explain how it can be progressed. [6M]
(c) Differentiate forward state-space search and backward state-space search. [6M]
- 5(a) Differentiate Passive Sensors and Active Sensors. [6M]
(b) List and briefly write about application domains of Robots. [6M]

(OR)

- (c) Explain the application and working of Ant colony system. [12M]

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M.Tech I Semester Regular/~~Supplementary~~ Examinations

MTIT1063-DIGITAL IMAGE PROCESSING
(Software Engineering (IT))

10 APR 2015

Time : 3 hours

Max.Marks :60

Answer all the questions
All Questions carry equal Marks

- 1(a) Explain about the basic relationships between pixels? [12M]
- (OR)**
- (b) Describe in detail about the components used in general purpose digital image processing system? [12M]
- 2(a) What effects would setting to zero the lower order bit-planes have on the histogram of an image in general? [6M]
- (b) Discuss different possible Low pass filtering masks. [6M]
- (OR)**
- (c) Describe in detail about power Linear transformation functions? [12M]
- 3(a) Describe how band reject and band pass filters are used for periodic noise reduction? [12M]
- (OR)**
- (b) Derive a CMY transformation to generate the compliment of a color image? [12M]
- 4(a) Explain the process of image segmentation using an global processing via Hough Transform? [12M]
- (OR)**
- (b) The images shown are quite different, but their histograms are identical. Suppose that each image is blurred with a 3X3 smoothing mask.
- a) Would the histogram is still identical after blurring [6M]
- b) If your answer is no, Sketch the two histograms? [6M]
- 5(a) Discuss the concept of matching by correlation? [6M]
- (b) Explain finite automata usage for string recognition? [6M]
- (OR)**
- (c) Specify the structure and weights of a neural network as a capable of performing exactly the same function as a minimum distance classifier for two pattern classes is n-dimensional? [12M]

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M.Tech I Semester Regular/~~Supplementary~~ Examinations

MTEC1061-BIO-MEDICAL SIGNAL PROCESSING

(Systems and Signal Processing (ECE))

10 APR 2015

Time : 3 hours

Max.Marks :60

Answer all the questions
All Questions carry equal Marks

- 1(a) Describe the method of representing a digital signal in frequency domain. Also discuss the properties of it. [7M]
(b) What is the necessity of FFT? Compare it over DFT. [5M]

(OR)

- (c) Discuss the decimation in time algorithm for N-point DFT with necessary equations. Also draw the flow graph for it. [12M]
2(a) Discuss the different kinds of transients in EEG and compare them. [6M]
(b) Explain the dynamics of sleep wake transitions. [6M]

(OR)

- (c) Discuss the markov model and draw the state diagram for a markov chain and explain. [12M]
3(a) Describe the recovery process of pitch of a vowel sound 'ah' in word **father**. [4M]
(b) Discuss the method of phase unwrapping using adaptive numerical integration. [8M]

(OR)

- (c) What is a cepstrum? Explain how to obtain the power cepstrum. [6M]
(d) What is phase unwrapping. [6M]
4(a) Discuss the adaptive noise cancelling with LMS algorithm and role of a convergence parameter ' μ '. [12M]

(OR)

- (b) Explain the Adaptive zero tracking of EEG signal for detecting epileptic patients. [12M]
5. Compare the following.
(a) LMS and RLS methods. [6M]
(b) Autocorrelation and burg methods. [6M]

(OR)

- (c) Describe the burg method to estimate the reflection coefficient and discuss how the forward and backward prediction errors are calculated using lattice filter. [12M]

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M.Tech I Semester Regular/~~Supplementary~~ Examinations

MTEE1061-INDUSTRIAL ELECTRONICS

(Power Electronics and Drives (EEE))

10 APR 2015

Time : 3 hours

Max. Marks :60

Answer all the questions
All Questions carry equal Marks

- 1(a) What is servo system? Explain it with the help of a block diagram. [6M]
(b) Describe the working principle of the UPS. [6M]
(OR)
(c) Briefly explain with the help of block diagrams the operation of Line interactive UPS. [6M]
(d) Briefly explain with the help of a simplified block diagram the operation of an ON – line UPS. [6M]
2. Give the construction and describe the working of
(a) Simple bimetal strip timer [6M]
(b) Dual bimetal strip timer. [6M]
(OR)
(c) State the need of a DC amplifier in industrial applications. [6M]
(d) Draw the Circuit and describe the operation of Loftin-white coupled dc amplifier. [6M]
- 3(a) What is an opto-coupler? Explain principle and operation of opto-coupler [6M]
(b) Define the following
i)Emitters ii)Sensors iii)Photo emissive cell iv)photo conductive cell
v)photo voltaic cell vi)photo junction devices. [6M]
(OR)
(c) When the wavelength of incident radiation on a photon – electric emitter is changed from 5000 to 6000 $\text{\AA}$. The maximum kinetic energy of the emitted electrons reduces by 20%. Calculate the work function of the emitter and the maximum kinetic energies in the two cases? [12M]
- 4(a) Describe with the help of a simple block diagram the principle of operation of a position control servo system. [6M]
(b) Mention the advantages of A.C. motor control in industry over the D.C. motor control. [6M]
(OR)
(c) Describe with the help of a circuit diagram the working of variable reluctance stepper motor. [12M]
- 5(a) Briefly describe the following applications of induction heating:
i) Surface hardening of steel ii) Annealing of brass and bronze items
iii) Brazing. [12M]
(OR)
(b) Compare dielectric heating with induction heating. [6M]
(c) Describe the principle of induction heating. [6M]

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M.Tech I Semester Regular/~~Supplementary~~ Examinations

MTME1061-ADVANCED POWER PLANT ENGINEERING

(Thermal Engineering (ME))

Time : 3 hours

Max.Marks :60

Answer all the questions
All Questions carry equal Marks

10 APR 2015

- 1(a) Do you think the power produced now is sufficient for the future generations? If so, for how many years and why you think so? If not, what alternatives do you suggest? [6M]
- (b) What is meant by power plant economics? What are fixed and operating costs? [6M]
- (OR)
- (c) Distinguish between commercial and non-commercial sources of energy. [6M]
- (d) Why non-conventional sources are considered to be future major power sources to overcome the power crisis? Which of these non-conventional sources do you think more prominent? Why? [6M]
- 2(a) What are the different types of coal conveyors? Sketch them and indicate the use of each and justify that it is essential at that stage. [12M]
- (OR)
- (b) With the help of line diagrams or sketches describe various ash handling systems. Give their relative merits and demerits. [6M]
- (c) With the help of a neat sketch, explain the working of governing system used for a closed-cycle gas turbine power plant. [6M]
- 3(a) What are the merits and demerits of using air or oxygen in gasification plant when integrated with closed cycle? [6M]
- (b) What is the importance of gasification of coal? Sketch such integration and explain its working. [6M]
- (OR)
- (c) Why Rankine cycle is preferred in waste heat recovery systems? What is the importance of fluid used in waste heat recovery? [12M]
- 4(a) What is moderator and what are the desirable properties of a moderator? Compare H_2O , D_2O and C as moderators. [6M]
- (b) What is fission? Explain with examples [6M]
- (OR)
- (c) Explain different types of nuclear reactions? Explain their significance in nuclear power generation. [6M]
- (d) Write a note on nuclear waste disposal system. [6M]
- 5(a) Enumerate the methods of detection or measurement of fuel leakages in a nuclear power plant [6M]
- (b) Explain the working of Orsat apparatus. [6M]
- (OR)
- (c) Explain with neat sketch the principle of working of McLeod gauge [6M]
- (d) What are thermocouples? How are they used to measure temperature. [6M]